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5g and 6g

harsha.m

untitled.m

untitled2.m

Workspace

Name	Value	Size
d	1×50 double	1×50
f	[900,3500,2...	1×3
Freq	[900,3500,2...	1×3
Loss	[91.5249,10...	1×3
PL1	1×50 double	1×50
PL2	1×50 double	1×50
PL3	1×50 double	1×50

untitled3 * ×

untitled4 * ×

untitled5 * ×

+

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26

PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);

PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on

plot(d,PL2,'r','LineWidth',2)

plot(d,PL3,'k','LineWidth',2)

title('Propagation Loss vs Distance')

xlabel('Distance (km)')

ylabel('Loss (dB)')

grid on

% ----- Graph 2 (NO legend used to avoid error) -----

subplot(2,1,2)

Freq = [900 3500 28000];

Loss = [PL1(10) PL2(10) PL3(10)];

plot(Freq,Loss,'-o','LineWidth',2)

title('Propagation Loss vs Frequency')

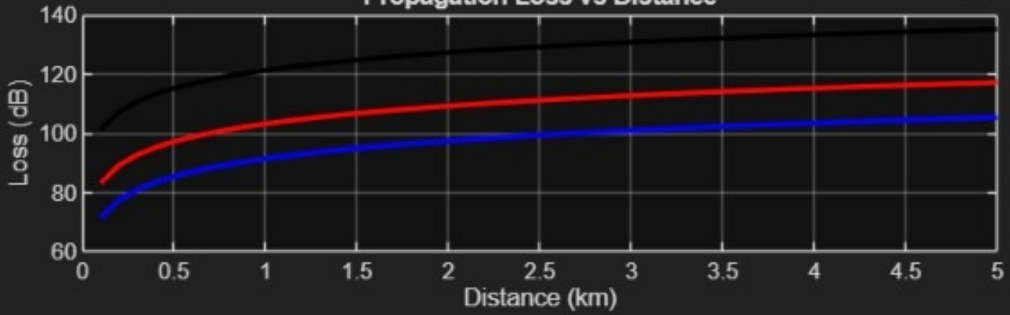
xlabel('Frequency (MHz)')

ylabel('Loss (dB)')

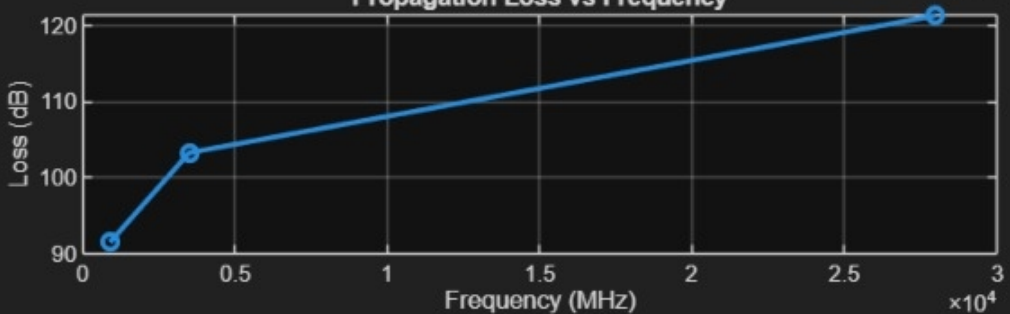
grid on

Figure 1 ×

Propagation Loss vs Distance



Propagation Loss vs Frequency



Command Window

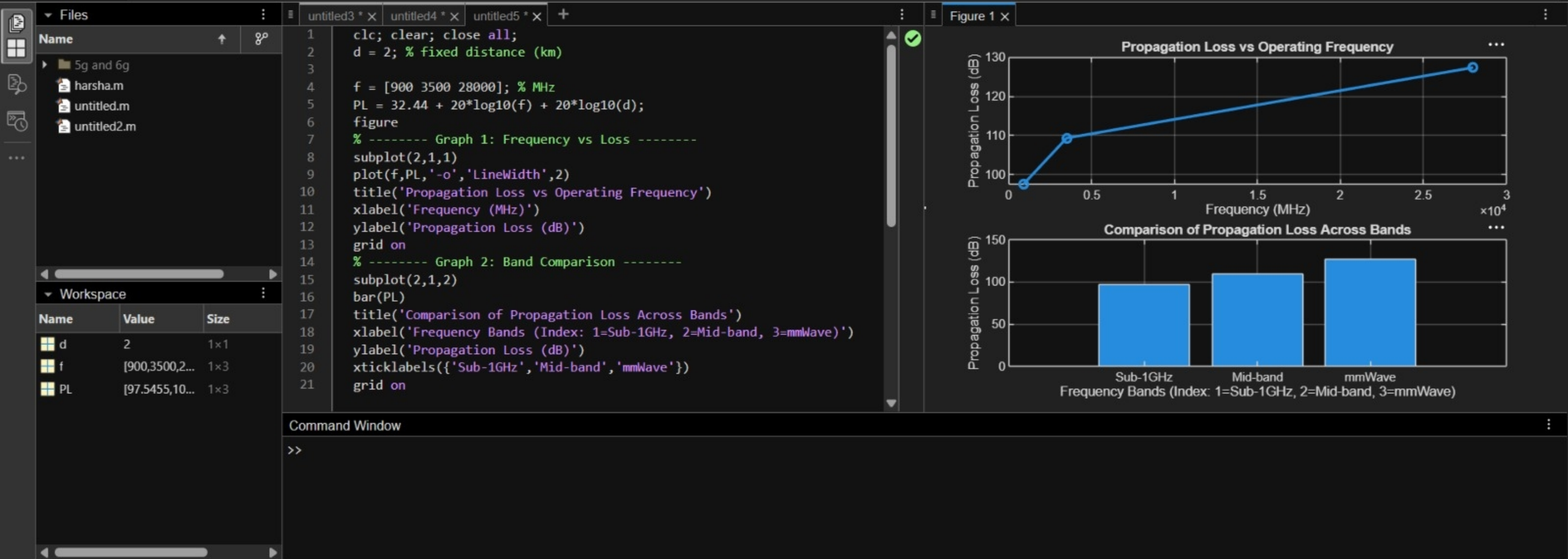
>>

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FILE

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untitled2.m

Workspace

Name	Value	Size
d	2	1×1
f	[900,3500,2...]	1×3
PL	[97.5455,10...]	1×3

untitled3 * ×

untitled4 * ×

untitled5 * ×

```
1 clc; clear; close all;
2 % Frequency bands (MHz)
3 f = [900 3500 28000];
4 % Distance (km)
5 d = 2;
6 % Propagation Loss (FSPL)
7 PL = 32.44 + 20*log10(f) + 20*log10(d);
8 figure
9 % ----- Graph 1: Simple Line Plot (SAFE) -----
10 subplot(2,1,1)
11 plot(f, PL, '-o', 'LineWidth', 2)
12 title('Propagation Loss vs Operating Frequency')
13 xlabel('Frequency (MHz)')
14 ylabel('Propagation Loss (dB)')
15 grid on
16 % ----- Graph 2: Safe Comparison Plot -----
17 subplot(2,1,2)
18 plot(1:3, PL, '-s', 'LineWidth', 2)
19 title('Band-wise Propagation Loss Comparison')
20 xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
21 ylabel('Propagation Loss (dB)')
22 grid on
```

Figure 1 ×

Propagation Loss vs Operating Frequency

Frequency (MHz)	Propagation Loss (dB)
900	97.5455
3500	109.5455
28000	127.5455

Band-wise Propagation Loss Comparison

Band Index	Propagation Loss (dB)
1	97.5455
2	109.5455
3	127.5455

Command Window

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